Editors' Suggestion

# Fermion parity and quantum capacitance oscillation with partially separated Majorana and quasi-Majorana modes

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In a recent experiment, flux-dependent oscillations of the quantum capacitance were observed in a one-dimensional spin-orbit-coupled semiconductor-superconductor heterostructure connected end to end via a quantum dot and threaded by a magnetic flux. In the topological superconducting phase hosted by this type of heterostructure, the oscillations corresponding to different fermion parity sectors are shifted by half a period and can serve as a mechanism for fermion parity readout or fusion operations involving a pair of localized, well-separated Majorana modes. In this work, we demonstrate that flux-induced fermion parity-dependent oscillations of the quantum capacitance in a disordered semiconductor–superconductor–quantum dot system can originate not only from topologically protected, spatially well-separated Majorana zero modes localized at the wire ends but also, generically, from partially separated Majorana modes with significant overlap, as well as from quasi-Majorana modes in the topologically trivial phase, which can be viewed as Andreev bound states whose constituent Majorana wave functions are slightly shifted relative to each other and have nonzero amplitude at opposite ends of the wire. Therefore, while the detection of flux-dependent oscillations of quantum capacitance marks an important experimental advance, such observations alone do not constitute conclusive evidence of the presence of topological Majorana zero modes.

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## I. INTRODUCTION

Majorana zero modes (MZMs) [1–3], the condensed matter counterparts of Majorana fermions from high-energy physics [4], can be understood as resulting from a nonlocal fractionalization of a conventional fermion mode into a pair of Majorana zero modes [1,2]. Due to their capacity to encode quantum information nonlocally and their associated exotic statistics, known as non-Abelian statistics [1,5–7], they have been proposed as fundamental building blocks for topological quantum computation [2,3,8]. It has been proposed that MZMs can be realized experimentally in a one-dimensional semiconductor-superconductor (SM-SC) heterostructure subjected to a parallel Zeeman field [9–12]. When the field exceeds a critical value, the system is predicted to transition into a one-dimensional topological superconducting (1DTS) phase that hosts MZMs localized at the ends, leading to extensive experimental efforts and several promising results in recent years [13–31]. However, it has also become evident that disorder can induce trivial low-energy states that often imitate the signatures of topological MZMs [32–47]. Notably, the so-called partially separated Andreev bound states (ABSs) or quasi-Majorana zero modes (q-MZMs) [38–40] are ABSs whose constituent Majorana wave functions are spatially separated just enough to form robust low-energy states resembling MZMs but not sufficiently to reduce their overlap to a level that would make the states insensitive to local perturbations, i.e., a separation that would provide topological protection. Such states are generically induced by strong-enough disorder in the topologically trivial phase of the heterostructure.

The clean or weakly disordered 1D SM-SC heterostructure with applied Zeeman field above a certain critical value is a system in which the lowest-energy excitations consist of a pair of MZMs localized at the wire ends. MZMs differ from conventional fermionic modes in that they do not possess a well-defined fermion occupation number. To define a conventional fermionic state in a system hosting MZMs, one must consider a pair of such modes. For example, the pair of MZMs $\gamma_1$ and $\gamma_2$, localized at the ends of a SM-SC hybrid nanowire in the topological phase, can be combined into a zero-energy complex fermion mode represented by the operator $c^\dagger = \frac{1}{2}(\gamma_1 + i\gamma_2)$. The charge state of the system is then given by the eigenvalue, 0 or 1, of $n_c = c^\dagger c$. The fermion parity $F = (-1)^{n_c}$ associated with the operator $c^\dagger$ is related to the MZM operators $\gamma_1$ and $\gamma_2$ via $F = (-1)^{n_c} = i\gamma_1\gamma_2$, indicating that the system's fermion parity is determined by nonlocal correlations between the spatially separated, fractionalized MZMs $\gamma_1$ and $\gamma_2$. In fact, this fractionalization of fermion parity into a pair of spatially separated MZM operators is a defining feature of the system's topological nature [48]. A measurement of the fermion parity shared by a pair of MZMs constitutes a fusion operation, which underlies measurement-only topological quantum computation involving Majorana qubits [49]. However, the nonlocal nature of the fermion parity, arising from its fractionalization between two spatially separated MZMs, renders its definitive measurement inherently challenging [50–52]. In a recent experiment [53], measurements of flux-dependent oscillations of the quantum capacitance [54–59] in a one-dimensional spin-orbit-coupled SM-SC hybrid nanowire connected end to end via a quantum

 

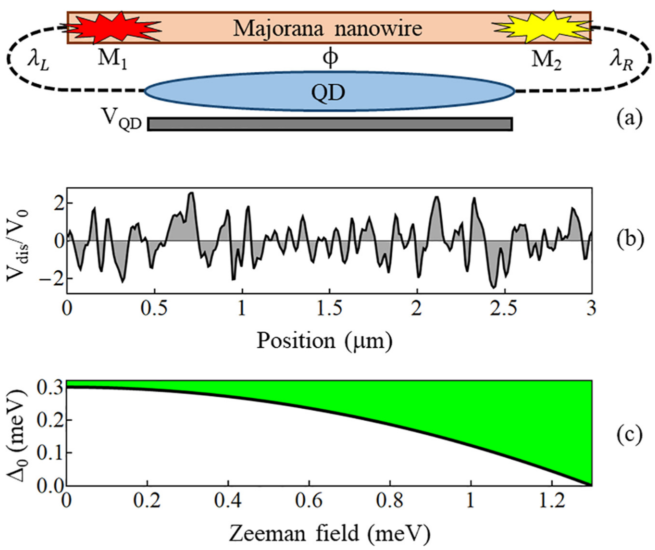


FIG. 1. (a) Schematic representation of a semiconductor-superconductor hybrid nanowire coupled to a quantum dot (QD). The effective coupling at the left (right) end of the nanowire is $t\lambda_L$ ($t\lambda_R$), where $t$ is the nearest-neighbor hopping of the 1D tight-binding model describing the SM wire [see text, Eqs. (2) and (5)]. A back gate controls the potential $V_{\rm QD}$ of the quantum dot, while a magnetic flux $\Phi$ threads the space between the wire and the dot. In the topological regime, the nanowire supports a pair of Majorana modes ($\mathrm{M}_1$ and $\mathrm{M}_2$) localized near the ends. (b) Position dependence of the (normalized) disorder potential in the nanowire. (c) Dependence of the parent superconducting gap on the Zeeman field applied parallel to the wire.

dot and threaded by a magnetic flux were used for fermion parity readout, or fusion operations involving a pair of Majorana modes, in a putative Majorana qubit.

In this paper, we study a one-dimensional SM-SC hybrid nanowire with Rashba spin-orbit coupling and a parallel magnetic field in the presence of intermediate to strong disorder. Reflecting the experimental setup, we assume that the two ends of the nanowire, referred to as a Majorana nanowire in Fig. 1, are connected end to end via an extended quantum dot (QD), forming a loop which is threaded by a perpendicular magnetic flux $\Phi$. In the topological superconducting phase of the SM-SC nanowire, the coupling of the end MZMs, $M_1$ and $M_2$, through the quantum dot gives rise to a pair of finite-energy ABSs, described by $c^\dagger|0\rangle = \frac{1}{2}(\gamma_1 + i\gamma_2)|0\rangle$ and $c|0\rangle = \frac{1}{2}(\gamma_1 - i\gamma_2)|0\rangle$, where $|0\rangle$ denotes the superconducting ground state. These states possess opposite fermion parities, $\pm 1$ (i.e., $i\gamma_1\gamma_2 c^\dagger|0\rangle = -c^\dagger|0\rangle$, $i\gamma_1\gamma_2 c|0\rangle = +c|0\rangle$), and differ in the occupation number of the conventional fermionic mode formed by $\gamma_1$ and $\gamma_2$ (i.e., $n_c = 1, 0$, respectively). The different occupation numbers of the ABS in the quantum dot result in distinct values of the quantum capacitance of the SM-SC-QD system. Therefore, measuring the quantum capacitance provides indirect access to the fermion parity associated with the end MZMs. The quantum capacitance of the QD may depend on factors other than the fermion parity (i.e., the ABS occupation), such as the flux $\Phi$ generated by a perpendicular magnetic field threading the interferometric loop, but this dependence is parity sensitive, providing a distinct signature of changes in the fermion parity [53].

We first calculate the flux dependence of the energies of the ABS pair ($c^\dagger|0\rangle$, $c|0\rangle$), corresponding to odd and even fermion parities, for a system with weak disorder. We show that analogous to the fractional Josephson effect [2], in the 1DTS phase of a weakly disordered 3 μm SM-SC quantum wire, the energy-flux relation $E(\Phi)$ exhibits periodicity with period $h/e$, which is twice the conventional superconducting flux quantum $h/2e$. Correspondingly, the quantum capacitance, computed using a linear response formalism [59], is also periodic in $\Phi$ with the same $h/e$ period. These flux-periodic oscillations of the quantum capacitance in the 1DTS phase with localized MZMs are consistent with recent experimental observations in QD-SM-SC heterostructures [53]. We then extend our analysis to the case of intermediate to strong disorder, again for a 3 μm quantum wire. Remarkably, we find that flux-dependent oscillations in fermion parity—or, equivalently, in quantum capacitance—can emerge not only from topologically protected MZMs localized at the ends of the wire but also, quite generically, from partially separated MZMs that exhibit significant spatial overlap while still maintaining nonzero amplitude at opposite ends of the wire. Even more significantly, we find that qualitatively similar parity-shifted, flux-dependent oscillations in capacitance can also arise from q-MZMs, which are generically induced by disorder in the topologically trivial phase. Consequently, while the observation of parity-dependent, flux-induced oscillations of the quantum capacitance in QD-SM-SC systems represents an important experimental advance, such oscillations alone do not constitute indisputable evidence of the presence of topological MZMs.

## II. HAMILTONIAN FOR THE HYBRID NANOWIRE COUPLED TO A QUANTUM DOT

We consider an interferometric loop consisting of a 3 μm Majorana nanowire, e.g., a quasi-1D SM-SC hybrid structure, coupled to a quantum dot (e.g., a SM wire) [53]. The device is represented schematically in Fig. 1(a). A magnetic field $B$ applied parallel to the wire generates a Zeeman splitting $\Gamma = \frac{1}{2}g\mu_B B$, where $g$ is the Landé $g$ factor of the semiconductor and $\mu_B$ is the Bohr magneton, while a magnetic flux $\Phi$ threads the interferometric loop. The SM wire is modeled using a simple 1D tight-binding Hamiltonian, and the proximity effect induced by the SC is captured, after integrating out the SC degrees of freedom, by a self-energy contribution. Specifically, the Green's function of the semiconductor has the form

$$G_{\rm SM}(\omega) = [\omega - H_{\rm SM} - \Sigma_{\rm SC}(\omega)]^{-1}, \tag{1}$$

with $H_{\rm SM}$ being the Bogoliubov–de Gennes (BdG) matrix that corresponds to the second quantized Hamiltonian

$$\begin{aligned}\hat{H}_{\rm SM} = &\sum_{i,\sigma}\{-t(c_{i\sigma}^\dagger c_{i+1\sigma} + \text{H.c.}) + [V_{\rm dis}(i) - \mu]c_{i\sigma}^\dagger c_{i\sigma}\}\\ &+ \frac{\alpha}{2}\sum_i[(c_{i\uparrow}^\dagger c_{i+1\downarrow} - c_{i\downarrow}^\dagger c_{i+1\uparrow}) + \text{H.c.}]\\ &+ \Gamma\sum_i(c_{i\uparrow}^\dagger c_{i+1\downarrow} + c_{i\downarrow}^\dagger c_{i+1\uparrow}),\end{aligned} \tag{2}$$

where $i = 1, 2, \ldots, N$ (with $N = 300$) labels the sites of a 1D lattice with lattice constant $a = 10$ nm and $c^\dagger_{i\sigma}$ ($c_{i\sigma}$) designates the creation (annihilation) operator for an electron with spin $\sigma$ occupying site $i$. The effective parameters are the nearest-neighbor hopping $t = 16.56$ meV, which corresponds to an effective mass $m^* = 0.023m_0$ characterizing an InAs semiconductor wire, the chemical potential $\mu$, the Rashba spin-orbit coupling coefficient $\alpha = 1.4$ meV (i.e., $\alpha a = 140$ meV Å), and the Zeeman field $\Gamma$. We assume that the SM wire has disorder, which is described by the random potential $V_{\rm dis}$, with $\langle V_{\rm dis}\rangle = 0$ and $\langle V_{\rm dis}^2\rangle = V_0^2$. The specific disorder profile used in the numerical calculations is shown in Fig. 1(b). The self-energy describing the superconducting proximity effect is assumed to be local, with contributions of the form

$$[\Sigma_{\rm SC}(\omega)]_{ii} = -\gamma\frac{\omega + \Delta_0\sigma_y\tau_y}{\sqrt{\Delta_0^2 - \omega^2}}, \tag{3}$$

where $\sigma_y$ and $\tau_y$ are Pauli matrices associated with the spin and particle-hole degrees of freedom, respectively, $\Delta_0$ is the gap of the parent superconductor (e.g., Al), and $\gamma = 0.2$ meV is the effective SM-SC coupling. We assume that the SC gap decreases as a function of the applied field, $\Delta_0(\Gamma) = \Delta_0(0)(1 - \Gamma^2/\Gamma_c^2)$, as shown in Fig. 1(c), having a maximum value $\Delta_0(0) = 0.3$ meV and completely collapsing at $\Gamma_c = 1.25$ meV, which for typical values of the $g$ factor characterizing InAs wires ($g \sim 10$–$15$) corresponds to a magnetic field $B \sim 2.9$–$4.3$ T. Since we focus on the low-energy physics, e.g., on the Majorana modes that may emerge in the hybrid nanowire [see Fig. 1(a)], it is convenient to work in the so-called static approximation, $\sqrt{\Delta_0^2 - \omega^2} \approx \Delta_0(\Gamma)$, and define the effective BdG Hamiltonian

$$H_{\rm eff} = ZH_{\rm SM} + \mathbb{I}\,\Delta\sigma_y\tau_y, \tag{4}$$

where $Z = \Delta_0/(\Delta_0 + \gamma)$ is the quasiparticle residue, $\mathbb{I}_{ij} = \delta_{ij}$, and $\Delta = \Delta_0\gamma/(\Delta_0 + \gamma)$ is the induced pairing potential. Note that the quantities $Z$ and $\Delta$ depend on the applied Zeeman field (through $\Delta_0$), decreasing from the maximum values $Z(0) = 0.6$ and $\Delta(0) = 0.12$ meV that correspond to zero magnetic field and eventually vanishing at $\Gamma = 1.25$ meV. Also note that the "bare" parameters characterizing the SM Hamiltonian in Eq. (2) are renormalized in the effective Hamiltonian by a factor $Z(\Gamma)$.

As shown in Fig. 1(a), the hybrid nanowire is coupled to a QD characterized by a discrete set of states with energies $\epsilon_n + V_{\rm QD}$ that can be tuned using a gate potential. We assume that the energy spacing is large compared with the relevant energy scales, and for simplicity, we focus on a single-QD state $n_0$, with $\epsilon_{n_0} = 0$. The corresponding amplitudes of the left and right dot-nanowire couplings are $t\lambda_L$ and $t\lambda_R$, respectively, while the phases are controlled by the magnetic flux $\Phi$ threading the loop. Explicitly, the (second quantized) Hamiltonian describing the QD and the nanowire-QD coupling has the form

$$\begin{aligned}\hat{H}_{\rm QD} = &\sum_\sigma (V_{\rm QD} - \mu)a^\dagger_\sigma a_\sigma \\ &- t\sum_\sigma[(\lambda_L\, e^{i\varphi}c^\dagger_{1\sigma}a_\sigma + \text{H.c.}) \\ &+ (\lambda_R\, e^{-i\varphi}c^\dagger_{N\sigma}a_\sigma + \text{H.c.})],\end{aligned} \tag{5}$$

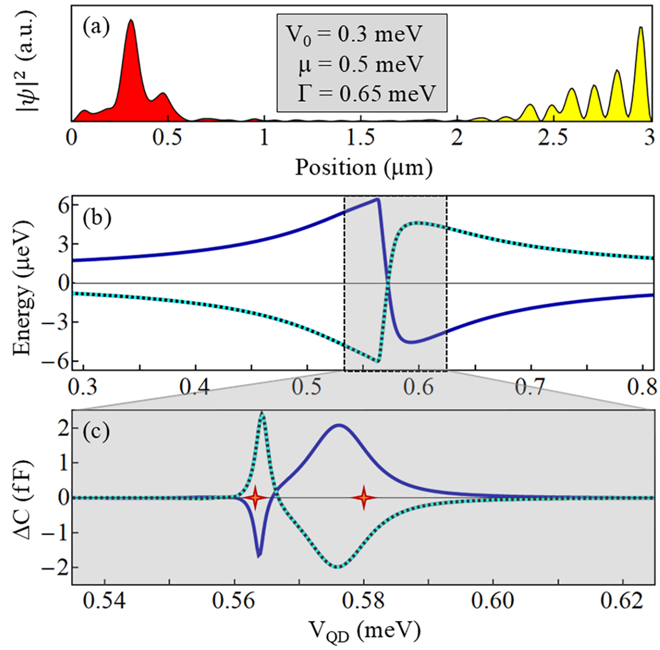


FIG. 2. (a) Spatial profiles of Majorana modes supported by a 3 μm long wire with low disorder of amplitude $V_0 = 0.3$ meV. The overlap of the two Majorana modes (see text) is $\mathcal{O} = 0.00056$. (b) Dependence of the lowest-energy mode (corresponding to a fixed parity) on the quantum dot potential for $\Phi = 0$ (dark blue line) and $\Phi = h/2e$ (dashed line). The disorder and control parameters are the same as in (a), and the coupling to the quantum dot corresponds to $\lambda_L = \lambda_R = 0.035$. Note the "resonance" (and the sign changes) corresponding to $V_{\rm QD} \gtrsim \mu$. (c) Capacitance difference between different parity states as a function of the quantum dot potential $V_{\rm QD}$ for $\Phi = 0$ (dark blue line) and $\Phi = h/2e$ (dashed line). The flux-dependent oscillations of the energy and the capacitance corresponding to the dot potential values marked by red crosses are shown in Fig. 3.

where $a^\dagger_\sigma \equiv c^\dagger_{0\sigma}$ ($a_\sigma \equiv c_{0\sigma}$) is the creation (annihilation) operator corresponding to the dot state $n_0$ with spin $\sigma$ and $\varphi = (\pi/2)\Phi/\Phi_0$, with $\Phi_0 = h/2e$, is a flux-dependent phase. Here, the notation $c_{0\sigma}$ corresponds to labeling the QD as site $i = 0$ of the 1D lattice.

## III. RESULTS

### A. Flux-dependent capacitance oscillation for topologically protected Majorana modes with negligible overlap

To benchmark our calculations, we first consider a hybrid nanowire with low disorder ($V_0 = 0.3$ meV) and tune the control parameters to a specific point within the topological phase corresponding to $\mu = 0.5$ meV and $\Gamma = 0.65$ meV. The lowest-energy state of the effective BdG Hamiltonian given by Eq. (4) corresponds to the pair of Majorana modes with spatial profiles given in Fig. 2(a). To characterize the spatial separation of the Majorana modes we define the "overlap" of the corresponding probability distributions,

$$\mathcal{O} = \frac{\sum_\alpha |\psi_{M_1}(\alpha)|^2|\psi_{M_2}(\alpha)|^2}{\sqrt{\sum_\alpha |\psi_{M_1}(\alpha)|^4\sum_\alpha |\psi_{M_2}(\alpha)|^4}}, \tag{6}$$

where $\alpha = (i, \sigma, \tau)$, representing position, spin, and particle-hole labels, takes $4N$ values and $\psi_{M_\ell}(\alpha)$ is the wave function of the $\ell$ Majorana mode. Note that $\mathcal{O} = 1$ corresponds to a pair of perfectly overlapping Majoranas, while $\mathcal{O} = 0$ indicates a pair of perfectly separated Majorana modes. The example shown in Fig. 2(a) corresponds to a pair of well-separated topologically protected Majoranas with $\mathcal{O} = 5.6 \times 10^{-4}$.

Next, we couple the hybrid wire to the quantum dot and diagonalize the corresponding BdG Hamiltonian, $H_{\rm eff} + H_{\rm QD}$. The low-energy states of the coupled system depend on the wire-QD couplings $\lambda_L$ and $\lambda_R$, the gate potential on the dot $V_{\rm QD}$, and the flux $\Phi$ that threads the interferometric loop. As an example, in Fig. 2(b) we show the dependence of the lowest-energy BdG mode on the QD potential $V_{\rm QD}$ for a system with $\lambda_L = \lambda_R = 0.035$ and magnetic flux $\Phi = 0$ (solid dark blue line) or $\Phi = h/2e$ (dashed line). Note that the two curves correspond to a state with well-defined (odd) parity; the state with opposite (even) parity has energy $E_e(V_{\rm QD}, \Phi) = -E_o(V_{\rm QD}, \Phi)$. The strongest dependence on the dot potential corresponds to a "resonance" near $V_{\rm QD}$ values slightly larger than the chemical potential. Within the resonance region, we calculate the zero-frequency capacitance for the odd and even parity sectors, which is given by the $\omega = 0$ value of the dynamical capacitance [53,59]

$$C(\omega) = \kappa \sum_{n,m}^{*} \frac{|\langle\psi_n|\tau_z\, \delta_{i0}|\psi_m\rangle|^2(E_n - E_m)}{(E_n - E_m)^2 - (\omega + i\eta)^2}, \tag{7}$$

where $\kappa = 0.032$ fF µeV and $\eta = 2$ µeV. The parameter $\eta$ accounts for the broadening of the resonance due to the coupling to the environment [53] and results in a suppression of the maximum values of the capacitance within the resonance region. Note, however, that the value of this parameter does not qualitatively affect the results. The matrix elements in the numerator are calculated using the BdG states $|\psi_n\rangle$ and $|\psi_m\rangle$ of energies $E_n$ and $E_m$, respectively. We calculate the capacitance for a system with either odd or even parity, which corresponds to the following constraints in the summation in Eq. (7). For the odd (even) capacitance, the index $n$, indicating empty states, takes the values $n \geqslant 2N + 2$ and $n = e$ $(o)$, while $m$, which indicates occupied states, takes the values $m \leqslant 2N - 1$ and $m = o$ $(e)$. Here, $e$ and $o$ designate the lowest-energy modes (i.e., modes $2N$ or $2N + 1$) having energies $E_e = -E_o$ and corresponding to even and odd parities, respectively. The presence of a constraint is indicated by the asterisk in Eq. (7). The capacitance difference between the even and odd sectors, $\Delta C = C_e - C_o$, as a function of the quantum dot potential $V_{\rm QD}$ is shown in Fig. 2(c) for a system with magnetic flux through the interferometric loop $\Phi = 0$ (solid dark blue line) and $\Phi = h/2e$ (dashed line). Note that $\Delta C$ has significant values (on the order of 1 fF) within a potential window of about 0.04 meV—the resonance region corresponding to $V_{\rm QD}$ values slightly above the chemical potential. Also note that $\Delta C$ changes sign at $V_{\rm QD} \approx 0.567$ meV, indicating a switch of the capacitance magnitudes characterizing the even and odd parity sectors. Finally, the signs of $\Delta C$ corresponding to $\Phi = 0$ and $\Phi = h/2e$ are typically opposite, which suggests that the capacitance values characterizing different parity sectors oscillate as a function of the applied magnetic flux.

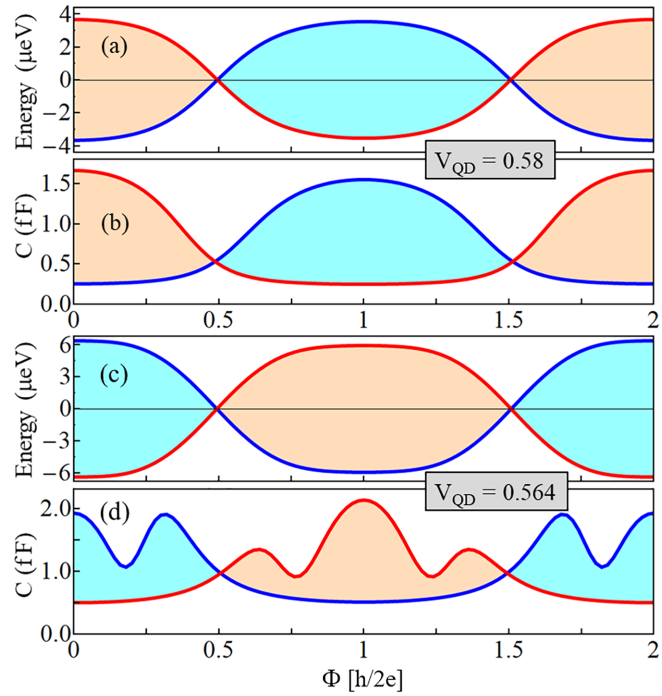


FIG. 3. Flux-induced oscillations of the energy and capacitance corresponding to different parity states of a (low-disorder) nanowire–quantum dot system with parameters as in Fig. 2. (a) Energy oscillations and (b) capacitance corresponding to a dot potential $V_{\rm QD}$ (given in meV) within the right resonance lobe in Fig. 2(c). (c) Energy and (d) capacitance as a function of the magnetic flux $\Phi$ for a dot potential within the left resonance lobe in Fig. 2(c).

The flux-induced oscillations of the capacitance and of the corresponding lowest-energy modes are illustrated in Fig. 3. We consider two values of the dot potential, one within the right "resonance lobe" ($V_{\rm QD} = 0.58$ meV) and the other within the left lobe [$V_{\rm QD} = 0.564$ meV; also see Fig. 2(c)]. For a given parity, the period of the oscillations is $h/e$. The amplitude of the energy oscillations, which is on the order of a few µeV, depends strongly on the coupling between the wire and the quantum dot, i.e., on the parameters $\lambda_L$ and $\lambda_R$. The capacitance oscillations have the same main features as the energy oscillations but can exhibit specific details, e.g., the presence of a higher oscillatory component in Fig. 3(d). The key point is that in the presence of weak disorder the hybrid Majorana wire supports well-separated Majorana modes [Fig. 2(a)], which in an interferometric device [Fig. 1(a)] lead to flux-induced oscillations of the capacitance corresponding to the odd and even parity sectors that are shifted relative to each other by half a period. The period of the oscillations (for a given parity) is $h/e$, and the maximum difference between the capacitance values corresponding to the two sectors is on the order of 1 fF.

### B. Flux-dependent capacitance oscillation for partially separated Majorana modes and quasi-Majorana modes

What is the corresponding picture in a system with intermediate to strong disorder? To address this question, we consider a system with the same parameters as those used

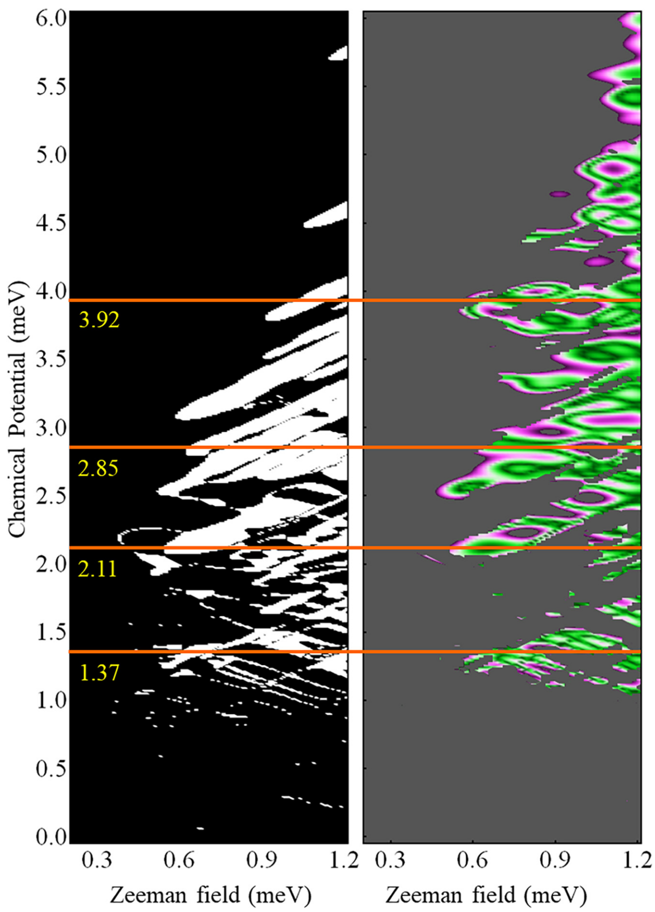


FIG. 4. Left: Topological phase diagram for a hybrid nanowire with intermediate to strong disorder ($V_0 = 1.2$ meV). The white and black regions are topologically nontrivial and trivial, respectively. The topological invariant was defined for an infinite system obtained by periodically repeating the $L = 3$ μm disordered wire (see text). Right: Low-energy map showing the control parameter region that supports delocalized states (see text) with energies between 0 μeV (dark green) and 10 μeV (pink). Parameter values within the dark gray region correspond to the lowest-energy states that are either localized or have energies larger than 10 μeV. Note the significant correlation between the left and right panels for chemical potential values below $\mu \approx 3$ meV. The low-energy spectra along the horizontal cuts marked by orange lines are shown in Fig. 5.

above, except the disorder amplitude, which is now $V_0 = 1.2$ meV. We first identify the control parameter regimes consistent with the presence of Majorana physics, e.g., islands within which the system could be considered to be in some operationally defined "topological" phase. The topological map calculated using the winding number invariant for the infinite system obtained by periodically repeating the finite disordered wire [60] is shown in Fig. 4 (left panel). Note that the topological region (white) is characterized by chemical potential values significantly larger than the $-1.2 \lesssim \mu \lesssim 1.2$ meV window corresponding to the clean system. In addition, the topological region is highly fragmented, indicating the presence of large finite size effects [60]. Relatively stable, near-zero-energy Majorana modes are expected to emerge within the large topological islands. We also map the lowest energy of the (finite) wire as a function of the Zeeman field and chemical potential under the constraint that the corresponding mode be delocalized. Our operational criterion for delocalization requires that 90% of the spectral weight of the lowest-energy BdG state be contained within a *connected* segment representing at least 85% of the wire length. We emphasize that the delocalized states consist of pairs of Majorana modes that, individually, could be either localized (e.g., at the ends of the wire) and characterized by an exponentially small overlap or rather extended, which would result in a large overlap. For example, a pair of "ideal" Majorana modes strongly localized at the ends of the wire and having nearly zero overlap would represent, according to this criterion, a maximally delocalized state that cannot be contained within a connected segment shorter than (almost) 100% of the wire length. The corresponding map is shown in the right panel of Fig. 4, with dark gray corresponding to either localized (lowest-energy) states or (gapped) states with energies larger than 10 μeV. The dark green regions indicate the presence of near-zero-energy (delocalized) modes. Note the significant correlation between the topological map (left) and the low-energy map (right) for chemical potential values $\mu \lesssim 3$ meV.

We now select four constant-$\mu$ cuts through parameter regions that support near-zero-energy (delocalized) modes, as indicated by the orange lines in Fig. 4. For $\mu \lesssim 3$ meV those regions correspond to topologically nontrivial islands (see Fig. 4), while the cut at $\mu = 3.92$ meV includes a relatively large Zeeman field window ($0.6 \lesssim \Gamma \lesssim 0.9$ meV) that supports a low-energy topologically trivial mode. As shown below, the constituent Majorana wave functions of this trivial mode are slightly displaced relative to each other, making it a partially separated ABS or a q-MZM. The energy spectra (as a function of the applied Zeeman field) corresponding to the four cuts are shown in Fig. 5. As expected [60], the (large) topological regions (gray shading) support relatively robust low-energy modes, with typical energy splittings of up to a few μeV. Of course, enhancing the robustness of these modes would require longer Majorana wires [60]. In addition, in Fig. 5(a) we show an example of a (relatively robust) topologically trivial low-energy mode which is a quasi-Majorana mode. Note that the near-zero-energy modes hosted by the low-field topological regions in Figs. 5(b) and 5(c) are protected by energy gaps of about 30 and 20 μeV, respectively, and that there is no signature corresponding to the closing of this gap at a topological quantum phase transition. This behavior indicates the presence of large finite-size effects [60]. The corresponding topological islands do not fulfill the topological gap protocol [31] employed to select experimental parameters for observing flux-dependent capacitance oscillations. Nonetheless, we find that a system with control parameters corresponding to the near-zero-energy modes (and nontrivial topological invariant values) shown in Figs. 5(b) and 5(c) exhibits flux-dependent oscillations in quantum capacitance similar to those observed experimentally, despite the absence of a closing and reopening of the bulk gap. Moreover, the system with parameters corresponding to Fig. 5(a) does so as well, even though it is topologically trivial.

The final step of the selection process involves choosing specific (nearly) zero-energy states. We focus on the low-field regime ($\Gamma \lesssim 0.8$ meV), which is more accessible and

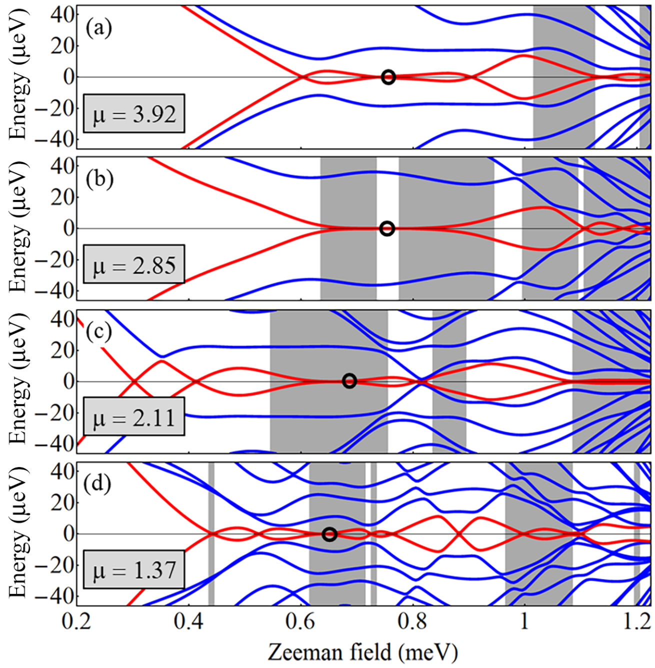


FIG. 5. Low-energy spectra as a function of the applied Zeeman field for a system with intermediate to strong disorder ($V_0 = 1.2$ meV) and different values of the chemical potential (given in meV) corresponding to the cuts marked in Fig. 4. The shaded regions correspond to a nontrivial value of the topological invariant. The spatial profiles of the Majorana modes corresponding to the near-zero-energy states marked by the black circles are shown in Fig. 6.

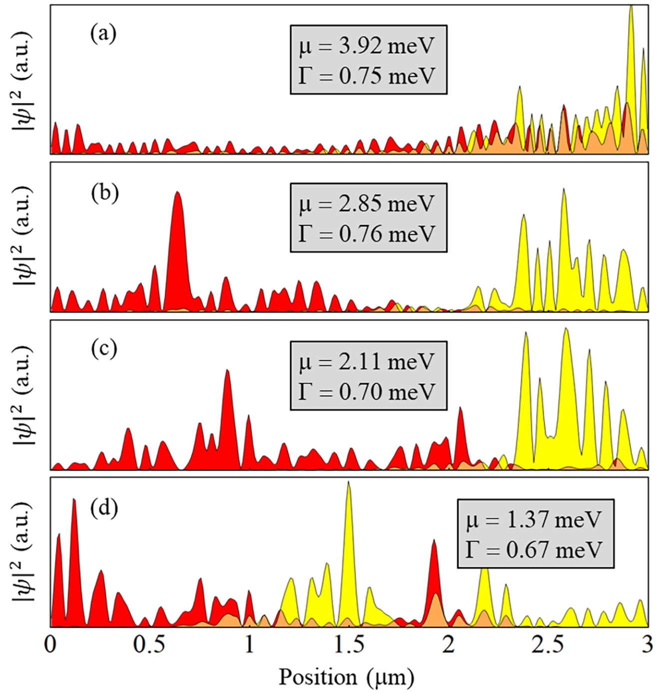


FIG. 6. Spatial profiles of the Majorana modes associated with the near-zero-energy states marked in Fig. 5. The control parameters are explicitly given in the corresponding panels. Note that all these states can be viewed as partially separated (or partially overlapping) Majorana modes. The corresponding overlaps, as defined in Eq. (6), are (a) $\mathcal{O} = 0.63$, (b) $\mathcal{O} = 0.046$, (c) $\mathcal{O} = 0.066$, and (d) $\mathcal{O} = 0.23$. The lowest overlap, corresponding to the modes in (b), is almost 2 orders of magnitude larger than the overlap characterizing the low-disorder modes in Fig. 2(a).

hence more relevant experimentally. The selected states are marked by black circles in Fig. 5, while the spatial profiles of the corresponding Majorana modes are shown in Fig. 6. The low-energy states correspond to pairs of partially separated Majorana modes with overlaps ranging from $\mathcal{O} = 0.046$ [Fig. 6(b)] to quasi-Majorana modes with $\mathcal{O} = 0.63$ [Fig. 6(a)]. Note that the lowest overlap ($\mathcal{O} = 0.046$) is about 2 orders of magnitude larger than the overlap characterizing the well-separated Majorana modes shown in Fig. 2(a), while the overlap in Fig. 6(d) is significantly larger than that in Fig. 6(b). This property is generic and is due to the fact that in a (strongly) disordered system the low-energy Majorana modes are not exponentially localized near the ends of the wire and have relatively large characteristic length scales ranging from a few microns up to tens of microns, depending on the specific system parameters (e.g., SM-SC coupling, disorder amplitude, etc.). In general, the values of the characteristic length scale increase with increasing disorder strength (as the topological phase shifts toward larger values of the chemical potential and Zeeman field [60,61]) or with decreasing effective SM-SC coupling (which implies lower values of the renormalized effective mass, Fermi velocity, etc.). We point out that the system also supports strongly localized low-energy states (typically at low values of the chemical potential), but these states are topologically trivial (see the topological map in Fig. 4). Thus, the generic delocalized low-energy states emerging in (strongly) disordered Majorana hybrid wires of lengths on the order of a few microns consist of partially separated Majorana modes [Figs. 6(b)–6(d)], which are characterized by typical overlap values significantly larger than those associated with their clean system counterparts [Fig. 2(a)], and topologically trivial quasi-Majorana modes [Fig. 6(a)], which have overlaps of order 1. We emphasize that labeling some of these modes, e.g., those shown in Figs. 6(b) and 6(c), as topological Majorana zero modes is a matter of taste (i.e., dependent on the operational definition of "topology"), but in general, relatively robust low-energy q-MZMs—near-zero-energy modes consisting of a pair of strongly overlapping Majoranas and having energy splittings smaller than (but comparable to) the experimental energy resolution over control parameter ranges (i.e., Zeeman field and chemical potential) on the order of the induced gap—exist that are definitely topologically trivial, e.g., the state shown in Fig. 6(a).

Having identified a set of representative (delocalized) near-zero-energy modes hosted by the Majorana hybrid wire with intermediate to strong disorder, we calculate the quantum capacitance signatures associated with different parity sectors using the interferometric setup consisting of the Majorana wire coupled to a quantum dot (see Fig. 1). The dependence of the capacitance difference between the even and odd parity sectors on the quantum dot potential is shown in Fig. 7. Figures 7(a)–7(d) show $\Delta C$ for a system with magnetic flux $\Phi = 0$ (dark blue lines) and $\Phi = h/2e$ (dashed lines) and control parameters (i.e., Zeeman field and chemical potential)

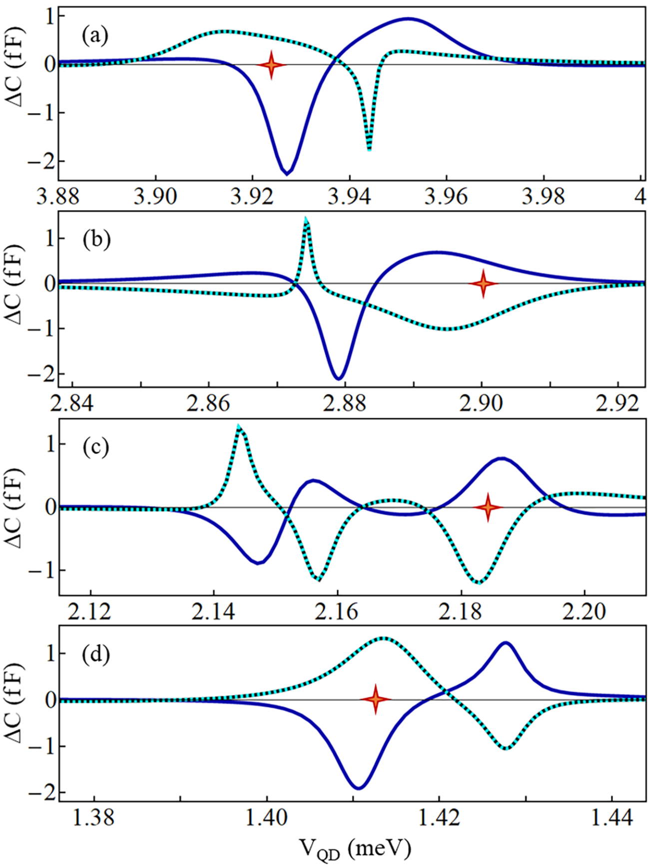


FIG. 7. Capacitance difference between different parity states as a function of the quantum dot potential $V_{\mathrm{QD}}$ for a nanowire–quantum dot system with control parameters given in the corresponding panels of Fig. 6. The dark blue lines are for a system with $\Phi = 0$, while the dashed lines correspond to $\Phi = h/2e$. The flux-dependent oscillations of the capacitance corresponding to the dot potential values marked by red crosses are shown in Fig. 8. The nanowire–quantum dot coupling parameters and the specific $V_{\mathrm{QD}}$ values corresponding to the red crosses are (a) $\lambda_L = 0.02$, $\lambda_R = 0.01$, and $V_{\mathrm{QD}} = 3.922$ meV; (b) $\lambda_L = 0.025$, $\lambda_R = 0.025$, and $V_{\mathrm{QD}} = 2.9$ meV; (c) $\lambda_L = 0.035$, $\lambda_R = 0.03$, and $V_{\mathrm{QD}} = 2.185$ meV; and (d) $\lambda_L = 0.025$, $\lambda_R = 0.025$, and $V_{\mathrm{QD}} = 1.413$ meV.

given in the corresponding panels of Fig. 6. Note that $\Delta C$ has significant values (on the order of 1 fF) within narrow resonance regions corresponding to $V_{\mathrm{QD}}$ values slightly above the chemical potential and changes sign at least once, qualitatively similar to the ideal case illustrated in Fig. 2.

Finally, we calculate the dependence of the capacitance characterizing the odd and even parity sectors on the magnetic flux $\Phi$ for specific values of the quantum dot potential marked by red crosses in Fig. 7. The parity-dependent oscillations shown in the corresponding panels of Fig. 8 represent the main result of this work. For a given parity, the period of these oscillations is $h/e$, while their amplitude is on the order of 1 fF, similar to the ideal (low-disorder) scenario illustrated in Fig. 3. We emphasize that there are no qualitative (or even significant quantitative) differences among the four panels in Fig. 8 or between these cases and the ideal scenario shown in Fig. 3. In other words, parity-selective oscillations of the

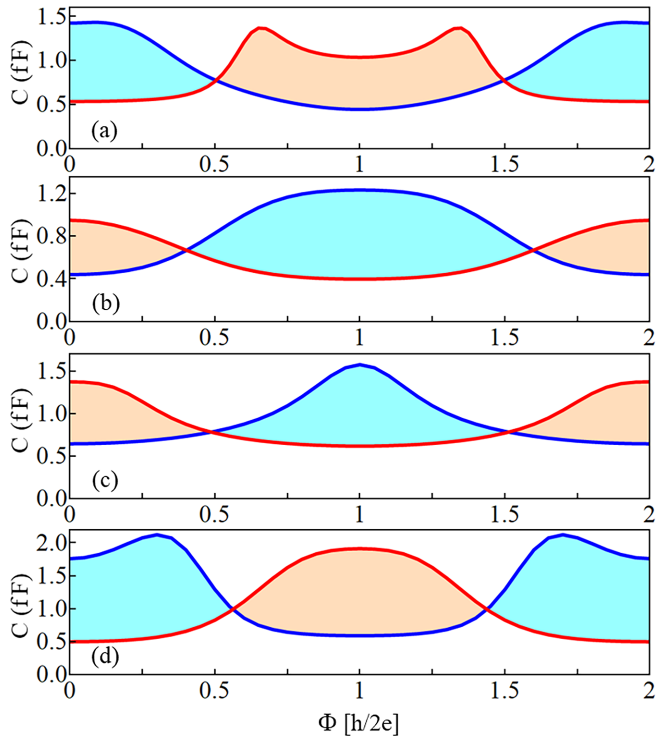


FIG. 8. Flux-dependent oscillations of the capacitance corresponding to the control parameter values given in Fig. 6 and the quantum dot potential values indicated in Fig. 7. Note that the main features and even some quantitative aspects (e.g., the amplitude of the oscillations) are basically independent of the underlying low-energy states and are similar to those characterizing the ideal case shown in Fig. 3.

capacitance (having period $h/e$) measured in an interferometric device consisting of a disordered, few micron long Majorana wire coupled to a quantum dot can indicate the presence of partially separated and strongly overlapping Majorana modes and even topologically trivial quasi-Majorana modes [38–40]. However, these modes are not necessarily topologically protected (i.e., well separated) and may even emerge in a topologically trivial phase. The probability that the partially separated Majorana modes responsible for the conductance oscillations are topologically protected decreases strongly with the disorder strength.

## IV. CONCLUSION

Motivated by the interesting results of a recent experiment [53], we considered an interferometric device consisting of a hybrid semiconductor-superconductor nanowire coupled to a quantum dot and calculated the quantum capacitance of the dot as a function of the magnetic flux through the interferometric loop for control parameter values corresponding to representative low-energy states of the wire. The experiment reported a fermion parity-dependent shift of the quantum capacitance of up to 1 fF and the observation of flux $h/2e$-periodic bimodality, which is consistent with parity-selective capacitance oscillations of period $h/e$. We modeled the semiconductor wire using a simple tight-binding Hamiltonian and incorporated the proximity effects induced by the parent superconductor through a self-energy term. The effective model parameters are consistent with the values characterizing the InAs-Al hybrid devices used in the experiment.

After benchmarking our quantum capacitance calculations using a low-disorder system that hosts a pair of nearly ideal Majorana zero modes, we focused on a hybrid nanowire with intermediate to strong disorder. First, we mapped the relevant control parameter space and showed that the generic delocalized low-energy states supported by the 3 μm long hybrid nanowire consist of partially separated Majorana modes. The characteristic length scales of these modes are on the order of 1–3 μm, comparable to the length of the system, and their overlap is at least 2 orders of magnitude larger than the overlap characterizing the ideal pair of Majoranas hosted by the low-disorder wire. Furthermore, while some of these Majorana modes could be dubbed "topological" (within a relaxed operational definition of "topology"), others are clearly topologically trivial quasi-Majorana modes with overlaps of order 1. We emphasize that, for the given values of the system parameters, enhancing the likelihood of realizing topological Majorana modes (i.e., well-separated modes with low overlap values) requires reducing the disorder strength. However, this would lead to the emergence of large topological regions (in the control parameter space), instead of separated topological islands, which appears to be inconsistent with the experimental situation. On the other hand, increasing the disorder strength would enhance the fragmentation of the topological phase and would increase the (typical) values of the overlap and the likelihood of having (relatively robust) nearly zero trivial states.

The main result of our work, which is illustrated in Fig. 8, concerns the emergence of parity-dependent oscillations of the quantum capacitance of the dot as a function of the magnetic flux threading the interferometric loop. These oscillations, of period $h/e$ and amplitudes on the order of 1 fF (consistent with recent experimental observations), can arise not only from exponentially localized Majorana modes with very little overlap, which emerge in the topologically nontrivial phase of weakly disordered system (see, e.g., Fig. 2), but also from near-zero-energy extended states consisting of a pair of partially overlapping Majorana modes with characteristic lengths of the order of the wire length. Such extended and partially overlapping Majorana modes can appear in the so-called topological patch regions of the control parameter space characterized by nontrivial values of a topological invariant [see Figs. 5(b)–5(d) and 6(b)–6(d)], consistent with previous results [59] and also in the topologically trivial phase as quasi-Majorana modes [38,40] having overlaps of order 1 [Figs. 5(a) and 6(a)]. These large overlap values, e.g., 65% in Fig. 6(a), do not affect the main features of the parity-dependent oscillations. We emphasize that the key requirement for observing parity-dependent capacitance oscillations is the existence of a low-energy delocalized BdG state consisting of a pair of partially overlapping Majorana modes. Hence, such oscillations are expected in all green regions in Fig. 4 (right panel). Note that, in practice, only the low-field regions (e.g., $\Gamma < 0.8$–1 meV) may be experimentally accessible. In the presence of weaker disorder, the states responsible for the capacitance oscillations are well-separated Majorana modes. However, as the disorder strength increases, quasi-Majorana-induced oscillations may become prevalent, and eventually, flux-induced capacitance oscillations may completely disappear in the low-field regime. It is encouraging that flux-induced capacitance oscillations have been observed experimentally, as that implies that the disorder strength characterizing the actual system is not significantly larger than the values used in our calculation. Nonetheless, our conclusion is that, in general, the observation of such oscillations indicates the presence of partially separated Majorana modes or topologically trivial quasi-Majorana modes and does not represent a unique signature of topologically protected Majorana zero modes.

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## DATA AVAILABILITY

The data that support the findings of this article are openly available [62], embargo periods may apply.

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